

Michael D. Bond

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RESEARCH INTERESTS	Developing analyses and systems that make complex, concurrent software reliable, scalable, and secure. Programming languages, runtime systems, compilers, security.	
EDUCATION	Ph.D., Computer Sciences, The University of Texas at Austin, 2008 – Title: Diagnosing and Tolerating Bugs in Deployed Systems – Advisor: Kathryn S. McKinley M.C.S., Computer Science, University of Illinois at Urbana-Champaign, 2003 B.S., Computer Science, University of Illinois at Urbana-Champaign, 2002	
AWARDS	ACM SIGPLAN Outstanding Doctoral Dissertation Award (2008) Intel Ph.D. Fellowship (2006–2008) Craziest idea (out of 10), Wild and Crazy Ideas session, ACM SIGPLAN International Symposium on Memory Management (ISMM) (2008) Best Student Presenter, IEEE / ACM International Symposium on Code Generation and Optimization (CGO) (2004) Microelectronics and Computer Development (MCD) Fellowship, The University of Texas at Austin (2003–2004)	
REFEREED PUBLICATIONS	Michael D. Bond, Katherine E. Coons, and Kathryn S. McKinley. Pacer: Proportional Detection of Data Races. <i>ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)</i> , Toronto, June 2010. To appear. Michael D. Bond, Graham Z. Baker, and Samuel Z. Guyer. Breadcrumbs: Efficient Context Sensitivity for Dynamic Bug Detection Analyses. <i>ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)</i> , Toronto, June 2010. To appear. Indrajit Roy, Donald E. Porter, Michael D. Bond, Kathryn S. McKinley, and Emmett Witchel. Laminar: Practical Fine-Grained Decentralized Information Flow Control. <i>ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)</i> , pages 63–74, Dublin, June 2009. Michael D. Bond and Kathryn S. McKinley. Leak Pruning. <i>ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)</i> , pages 277–288, Washington, DC, March 2009. Michael D. Bond and Kathryn S. McKinley. Tolerating Memory Leaks. <i>ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)</i> , pages 109–126, Nashville, TN, USA, October 2008. Michael D. Bond and Kathryn S. McKinley. Probabilistic Calling Context. <i>ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)</i> , pages 97–112, Montreal, October 2007.	

	Michael D. Bond, Nicholas Nethercote, Stephen W. Kent, Samuel Z. Guyer, and Kathryn S. McKinley. Tracking Bad Apples: Reporting the Origin of Null and Undefined Value Errors. <i>ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)</i>, pages 405–422, Montreal, October 2007. Byeongcheol Lee, Kevin Resnick, Michael D. Bond, and Kathryn S. McKinley. Correcting the Dynamic Call Graph Using Control Flow Constraints. <i>International Conference on Compiler Construction (CC)</i>, pages 80–95, Braga, Portugal, March 2007. Michael D. Bond and Kathryn S. McKinley. Bell: Bit-Encoding Online Memory Leak Detection. <i>ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)</i>, pages 61–72, San Jose, CA, USA, October 2006. Michael D. Bond and Kathryn S. McKinley. Continuous Path and Edge Profiling. <i>IEEE/ACM International Symposium on Microarchitecture (MICRO)</i>, pages 130–140, Barcelona, November 2005. Michael D. Bond and Kathryn S. McKinley. Practical Path Profiling for Dynamic Optimizers. <i>IEEE/ACM International Symposium on Code Generation and Optimization (CGO)</i>, pages 205–216, San Jose, CA, USA, March 2005. Rahul Joshi, Michael D. Bond, and Craig Zilles. Targeted Path Profiling: Lower Overhead Path Profiling for Staged Dynamic Optimization Systems. <i>IEEE/ACM International Symposium on Code Generation and Optimization (CGO)</i>, pages 239–250, Palo Alto, CA, USA, March 2004.
DISSERTATION	Michael David Bond. <i>Detecting and Tolerating Bugs in Deployed Systems</i>. Department of Computer Sciences, The University of Texas at Austin, December 2008. ACM SIGPLAN Outstanding Doctoral Dissertation Award.
TECHNICAL REPORT	Michael D. Bond, Varun Srivastava, Kathryn S. McKinley, and Vitaly Shmatikov. Efficient, Context-Sensitive Detection of Semantic Attacks. Technical report TR-09-14, The University of Texas at Austin, 2009.
WORK EXPERIENCE	Postdoctoral fellow at The University of Texas at Austin (2009–present) Research assistant at The University of Texas at Austin (2003–2008) Research intern at Intel Corporation in Hillsboro, Oregon (2005) Research assistant at University of Illinois at Urbana-Champaign (2003) Teaching assistant for Combinatorial Algorithms (CS 373) at University of Illinois at Urbana-Champaign (2002) Software engineering intern at Edge Technologies (summers 2000–2002) Assistant teacher for computer science course at Thomas Jefferson High School for Science and Technology (summers 1997–1999)
PROFESSIONAL ACTIVITIES	Program Committee, <i>ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)</i> 2010 External Review Committee, <i>ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)</i> 2010

	Program Committee, <i>ACM SIGSOFT International Workshop on Multicore Software Engineering (IWMSE)</i> 2010 Review Committee, <i>ACM SIGPLAN International Symposium on Memory Management (ISMM)</i> 2010 Reviewer for <i>ACM Transactions on Architecture and Code Optimization (TACO)</i>; <i>ACM Transactions on Software Engineering and Methodology (TOSEM)</i>; and <i>Software: Practice and Experience (SPE)</i> External reviewer for <i>ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)</i>; <i>ACM Symposium on Principles of Programming Languages (POPL)</i>; <i>ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)</i>; <i>IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)</i>; <i>IEEE/ACM International Conference on Parallel Architectures and Compilation Techniques (PACT)</i>; and <i>International Conference on Compiler Construction (CC)</i>
SERVICE & OUTREACH	Jikes RVM Team (2007–present) Faculty Recruiting Committee and Co-chair of Graduate Student Faculty Recruiting Committee, Department of Computer Sciences, The University of Texas at Austin (2007) System administrator for my research group’s experimental machines (2005–present) Taught chess to high schools students through the Boys and Girls Club (2006–2007) Tutored high school students in math through Austin Partners in Education (2005)
GRANT PROPOSALS	Participated in writing and research development for <i>PASS: Perpetually Available Software Systems</i> (NSF SHF-0910818; this grant currently funds me), <i>Scalable Applications Start with Scalable Virtual Machine Services</i> (NSF CSR-0917191), and <i>Efficient Dynamic Analysis for Detecting and Tolerating Program Anomalies</i> (NSF CCF-0811524)
STUDENTS CO-ADVISED	Varun Srivastava, master’s student (co-advised with Kathryn McKinley and Vitaly Shmatikov, 2008–present) Graham Baker, Ph.D. student, Tufts University (co-advised with Samuel Guyer, 2008–2009) Rudy Depena, master’s student (co-advised with Kathryn McKinley, 2008–2009) Stephen Kent, undergraduate student (co-advised with Kathryn McKinley, 2007–2008)
SOFTWARE ARTIFACTS	Laminar (with Indrajit Roy and Donald Porter), Jikes RVM Research Archive, http://www.jikesrvm.org/Research+Archive, 2009 Leak Pruning, Jikes RVM Research Archive, 2008 Bad Apples Suite, 12 real null pointer exception bugs (with Stephen W. Kent), http://www.cs.utexas.edu/~mikebond/bad-apples-suite, 2008 Melt: Memory Leak Tolerance, Jikes RVM Research Archive, 2008 Probabilistic Calling Context, Jikes RVM Research Archive, 2007 – Jones and Ryder used this implementation for context-sensitive allocation sites in their 2008 paper in <i>ACM SIGPLAN International Symposium on Memory Management (ISMM)</i>.

Origin Tracking for Null References, Jikes RVM Research Archive, 2007

- Co-author Nethercote implemented origin tracking for undefined values in Valgrind's Memcheck tool. It is available as a branch in the Valgrind Source Code Repository: [svn://svn.valgrind.org/valgrind/branches/ORIGIN_TRACKING](http://svn.valgrind.org/valgrind/branches/ORIGIN_TRACKING)

Sleigh: A Memory Leak Detection Tool, Jikes RVM Research Archive, 2006

- Tang, Gao, and Qin modified Sleigh for their 2008 paper in *USENIX Annual Technical Conference (USENIX)*.

PEP: Continuous Path and Edge Profiling, Jikes RVM Research Archive, 2006

Practical and Targeted Path Profiling, part of the Scale compiler, <http://www-ali.cs.umass.edu/Scale/>, 2005

- Vaswani, Nori, and Chilimbi modified the path profiling implementation for their 2007 papers in *ACM Symposium on Principles of Programming Languages (POPL)* and *ACM SIGSOFT Symposium on the Foundations of Software Engineering (FSE)*.

PRESENTATIONS

"Making Software Robust in Deployed Systems"

- University of Michigan at Ann Arbor, April 2009
- Duke University, March 2009
- University of Nebraska at Lincoln, February 2009

"Leak Pruning," *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, Washington, DC, March 2009

"Tolerating Memory Leaks," *ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*, Nashville, TN, USA, October 2008

"Deployed Software: An Ideal Environment for Fixing Bugs?"

- University of Illinois at Urbana-Champaign, July 2008
- University of Washington, June 2008
- University of Oregon, June 2008
- Stanford University, June 2008
- University of California at Santa Cruz, June 2008
- University of California at Berkeley, June 2008
- University of California at San Diego, June 2008
- Texas A & M University, June 2008

"Bounding Leaky Programs," Wild and Crazy Ideas Session, *ACM SIGPLAN International Symposium on Memory Management (ISMM)*, Tucson, AZ, USA, June 2008

- Awarded craziest idea (out of 10)
- Runner-up for most worthy of implementation

"Probabilistic Calling Context," *ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*, Montreal, October 2007

"Tracking Bad Apples: Reporting the Origin of Null and Undefined Value Errors," *ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*, Montreal, October 2007

"Improving Software Reliability at Production Time"

- IBM Research, Hawthorne, NY, USA, October 2007
- IBM Corporation, Austin, TX, USA, October 2007
- Intel Corporation, Hillsboro, OR, USA, June 2007

“Bell: Bit-Encoding Online Memory Leak Detection,” *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, San Jose, CA, USA, October 2006

“Continuous Path and Edge Profiling,” *IEEE/ACM International Symposium on Microarchitecture (MICRO)*, Barcelona, November 2005

“Practical Path Profiling for Dynamic Optimizers,” *IEEE/ACM International Symposium on Code Generation and Optimization (CGO)*, San Jose, CA, USA, March 2005

“Targeted Path Profiling: Lower Overhead Path Profiling for Staged Dynamic Optimization Systems,” *IEEE/ACM International Symposium on Code Generation and Optimization (CGO)*, Palo Alto, CA, USA, March 2004

– Best Student Presenter

REFERENCES

Available on request